

2024 级线性代数 I (A) 参考答案

一、1. 6 2. $\begin{pmatrix} 1/2 & 1/2 & 0 \\ -1/4 & 1/4 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}$ 3. 21 4. $\begin{pmatrix} 2025 & 2024 \\ -2024 & -2023 \end{pmatrix}$

5. 3 6. $-3 < t < 3$ 7. $\begin{pmatrix} -1 & 0 & -1 \\ 1 & -1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$

二、8. C 9. A 10. B 11. D 12. B

三、13. $(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{pmatrix} 2 & -1 & 3 & 8 \\ 1 & 1 & 1 & 5 \\ 1 & 7 & -1 & 9 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 5 \\ 2 & -1 & 3 & 8 \\ 1 & 7 & -1 & 9 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 5 \\ 0 & -3 & 1 & -2 \\ 0 & 6 & -2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 5 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
 $\sim \begin{pmatrix} 1 & 1 & 1 & 5 \\ 0 & 1 & -1/3 & 2/3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 4/3 & 13/3 \\ 0 & 1 & -1/3 & 2/3 \\ 0 & 0 & 0 & 0 \end{pmatrix},$ 4'

故 $R(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = 2$, 2'
 α_1, α_2 为一个极大无关组. 2'

$\alpha_3 = \frac{4}{3}\alpha_1 - \frac{1}{3}\alpha_2, \alpha_4 = \frac{13}{3}\alpha_1 + \frac{2}{3}\alpha_2.$ 4'

14. $|A| = \begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & 5-\lambda & -4 \\ -2 & -4 & 5-\lambda \end{vmatrix} = (\lambda-1)^2(10-\lambda).$ 4'

(1) 当 $\lambda \neq 1$ 且 $\lambda \neq 10$ 时, 方程组有唯一解. 2'

(2) 当 $\lambda = 10$ 时, $B = (A, b) = \begin{pmatrix} -8 & 2 & -2 & 1 \\ 2 & -5 & -4 & 2 \\ -2 & -4 & -5 & -11 \end{pmatrix} \sim \begin{pmatrix} 2 & -5 & -4 & 2 \\ 0 & -18 & -18 & 9 \\ 0 & -9 & -9 & 9 \end{pmatrix} \sim \begin{pmatrix} 2 & -5 & -4 & 2 \\ 0 & -9 & -9 & 9 \\ 0 & 0 & 0 & -9 \end{pmatrix},$

$R(A) = 2 \neq 3 = R(B)$, 方程组无解. 2'

(3) 当 $\lambda = 1$ 时, $B = (A, b) = \begin{pmatrix} 1 & 2 & -2 & 1 \\ 2 & 4 & -4 & 2 \\ -2 & -4 & 4 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$

$R(A) = R(B) = 1 < 3$, 方程组有无穷多解. 4'

通解为 $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = k_1 \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, k_1, k_2 \in \mathbf{R}$

$$15. |A - \lambda E| = \begin{vmatrix} 2-\lambda & 0 & 2 \\ 0 & 2-\lambda & 2 \\ 2 & 2 & 4-\lambda \end{vmatrix} = \lambda(\lambda-6)(2-\lambda).$$

$\therefore$ 特征值为 $\lambda_1 = 0, \lambda_2 = 2, \lambda_3 = 6$.

对 $\lambda_1 = 0$, 解方程组 $(A - \lambda_1 E)x = 0$:

$$A - \lambda_1 E = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 2 & 2 \\ 2 & 2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_1 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \Rightarrow p_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

对 $\lambda_2 = 2$, 解方程组 $(A - \lambda_2 E)x = 0$:

$$A - \lambda_2 E = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 2 \\ 2 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \Rightarrow p_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

对 $\lambda_3 = 6$, 解方程组 $(A - \lambda_3 E)x = 0$:

$$A - \lambda_3 E = \begin{pmatrix} -4 & 0 & 2 \\ 0 & -4 & 2 \\ 2 & 2 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & -1 \\ 0 & -2 & 1 \\ -2 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_3 = \begin{pmatrix} 1/2 \\ 1/2 \\ 1 \end{pmatrix} \Rightarrow p_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

$$\text{令 } P = (p_1, p_2, p_3) = \begin{pmatrix} -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{pmatrix}, A = \begin{pmatrix} 0 & & \\ & 2 & \\ & & 6 \end{pmatrix},$$

则 P 正交, 且 $P^{-1}AP = A$.

四、16. 显然 $B^T = B$.

$$\forall x \neq 0, x^T Bx = x^T (E + A^T A)x = x^T x + x^T A^T A x = x^T x + (Ax)^T (Ax) = \|x\|^2 + \|Ax\|^2 > 0,$$

故 B 为正定矩阵.

17. 设 $k_1 \alpha_1 + k_2 A(\alpha_1 + \alpha_2) = 0$,

则 $k_1 \alpha_1 + k_2 (\lambda_1 \alpha_1 + \lambda_2 \alpha_2) = 0$,

即 $(k_1 + \lambda_1 k_2) \alpha_1 + \lambda_2 k_2 \alpha_2 = 0$,

$\therefore \lambda_1, \lambda_2$ 是 A 的两个互异的特征值,

$\therefore \alpha_1, \alpha_2$ 线性无关,

于是 $\begin{cases} k_1 + \lambda_1 k_2 = 0, \\ \lambda_2 k_2 = 0, \end{cases}$

则 $k_1 = k_2 = 0 \Leftrightarrow \begin{vmatrix} 1 & \lambda_1 \\ 0 & \lambda_2 \end{vmatrix} \neq 0$, 即 $\lambda_2 \neq 0$.

故向量组 $\alpha_1, A(\alpha_1 + \alpha_2)$ 线性无关的充要条件是 $\lambda_2 \neq 0$.